Unit – IV

Digitial CMOS Circuits

Date
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Topics

- N-MOS, P-MOS, CMOS, MOSFET Para.
- Technology Scaling, Channel length modulⁿ.
- Hot electron effect, Velocity Satuⁿ
- CMOS, Inverter, Device, Sizing.
- CMOS combinational logic design
- Powes dissⁿ, Power-delay product.
- Body effect, Rise and fall times
- Latch up effect, Transⁿ gates

Technology Scaling & Scaling.

① Technology Scaling.

- Technology Scaling means reducing the size of Components like transistor.

- So the maximum number of devices can be placed.

- This improves Speed, Performance and efficiency of IC

- So many Times, it is required to change the Fabrication Technology or bias Vtg.

- Suppose we are using MOSFET transistors, and we want to accommodate a large number of transistors in a single IC package, then Scaling Technology is used.

* Types of Scaling.

① Constant field Scaling

- Constant field Scaling means reducing the transistors size and supply Vtg. by scaling factor

$$S = \frac{1}{\alpha}$$

- electric field inside the transistor remains constant.

② Constant Vtg Scaling.

- Same as reducing Transistor Size and supply Vtg remains unchanged

* Parameters.

Parameter	Scaling factor	Effect
① Gate Area [A]	$(1/\alpha) \times (1/\alpha) = \frac{1}{\alpha^2}$	Decreases
② Gate capacitance [Cg]	$\frac{1}{\alpha}$	Decreases
③ Gate density (Cd)	$\frac{1}{\alpha}$	Decreases
④ Power dissipation per Gate (P)	$\frac{1}{\alpha^2}$	Decreases

① Gate Area (A)
- The area occupied by the Transistor
$A = L \times W$
- Scaled by $(1/\alpha)^2$

② Gate Capacitance [Cg].
- Gate Capacitance scales by $1/\alpha$.
- It requires less charge to switch Transistor.

⑤ Gate Delay [τ_d]:

- Scale by $\frac{1}{\alpha}$.
- Speed is increase by α.

④ Power dissipation per Gate [P]

- The p.d scales by $1/\alpha^2$.

Adv

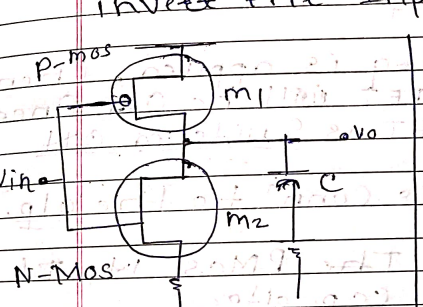
① Integ^n density (↓)
② Speed (↑)
③ Cost of chip (↓)

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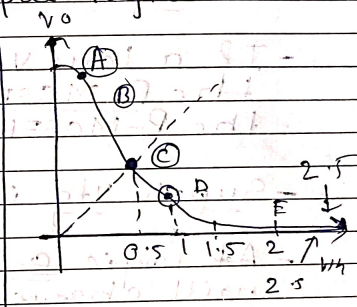
① Powee Consump^n (↑)
② Temp (↑).

CMOS Inveetee

- A CMOS inveetee is a basic buildin block in digitial circuits that invect the Input logic.



ckt diagram



Trans. chara

- In the inveetee, if you apply high I/p ; The o/p will be low
- If you apply low I/p , the o/p will be high.
- The inveeted using can be implemented by using Mosfet.
- There are two Types of mosfet is used
  $M_1 \rightarrow$ P-MOS.
  $M_2 \rightarrow$ N-MOS.

- M_1 and M_2 are Connected one after another.
- V_{in} is the input Vtg and V_o is the o/p voltage.
- If a low Vtg is applied, then the P-MOSFET will be ON. Once the P-MOSFET is switched ON
- Supply Vtg is Connⁿ to the o/p.
- At logic 0 The PMOS M_1 It will charge capacitor.
- At logic 1 [High i/p] M_1 is OFF and M_2 is ON.
- Then the Supply Vtg get Connected to the GND

Chara.

① Region A

- $V_{in} = 0$, NMOS = OFF, PMOS = ON

$0 < V_{in} < V_{th}$ ⟹ The Vtg. o/p is equal to V_{DD} at the o/p

② Region B

- M_1 is linear mode, M_2 goes into Saturation
- M_2 was in the Cut-off condⁿ.
- o/p vtg is (1).

③ Region C

- M_1 and M_2 are in Saturⁿ Mode.

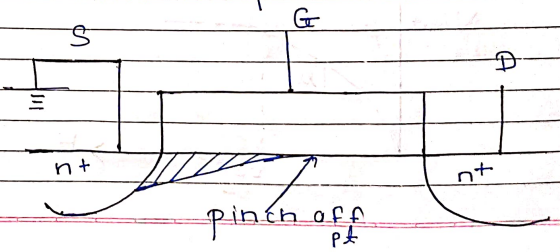
④ Region D

- M_1 is in Saturation & M_2 is in linear mode

⑤ Region E

- M_1 is OFF, M_2 is ON.
- discharging the capacitor takes place.

* Channel length modulation.



- The length of channel is change due to certain conditions is called Channel length modulation.

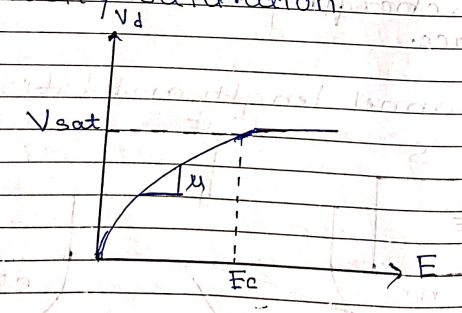
- In Saturation mode, transisters is acts as perfect current Source i.e. I_{DS} = Constant.

- not dependent on V_{DS}

- But practically, it's not true, if we increasing V_{DS}, the depletion region at drain junction is (↑) and effective length reduce.

- V_{DS} (↑) pinch-off point shift towards the Source terminal is called as pinch-off condition.

* Velocity Saturation.



- The Velocity (V_d) of a charged carrier is directly proportional to the applied electric field (E) and prop. Constant is μ.

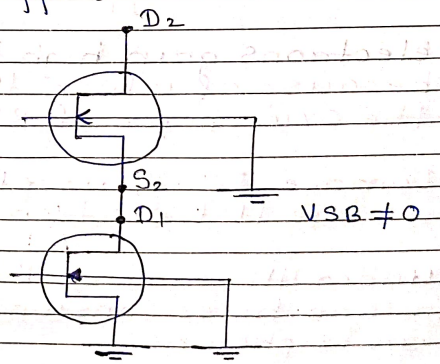
$$V_d \propto E$$

- μ → mobility of charged carrier
- The graph of V_d Vs E.

- when this value of E means electric field reaches a critical value E_c.

- The Velocity of a charged carrier changes or tends to Saturate and (μ) is decreases.

Body effect



- If there are Similar types of MOSFETS, i.e. P-Type & N-Type,

- These Similar Types of MOSFET are fabricated by using Common Substrate.

- V_{SB} → Source Bulk Vtg.

- for lower MOS, the S, terminal is directly conn to ground. So V_{SB} is zero.

- In in upward direc^n, $V_{SB} \neq 0$, means
- Value of Source to bulk is not kept at zero, It's changing.

* Hot electron effect

- electrons gain high kinetic energy because of which they penetrate the oxide layer of structure.

- Damage IC. This effect is known as the Hot electron effect.

- under thermal equilibrium, electrons and holes Constantly absorbs & emits photons.

- Avg gain in their KE is zero and Velocity (↑) as E (↑) [100 kv/cm].

- These Carriers does not remain in thermal eqlb.

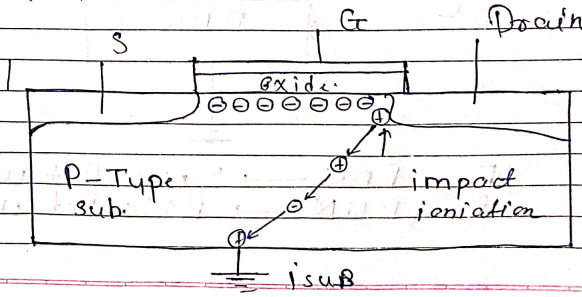
- Causes impact or Avalanche ionization. multip^n.

- These Carriers injects in oxide layer and may damage the IC.

* Protection
① Use add^n MOSFET in Series

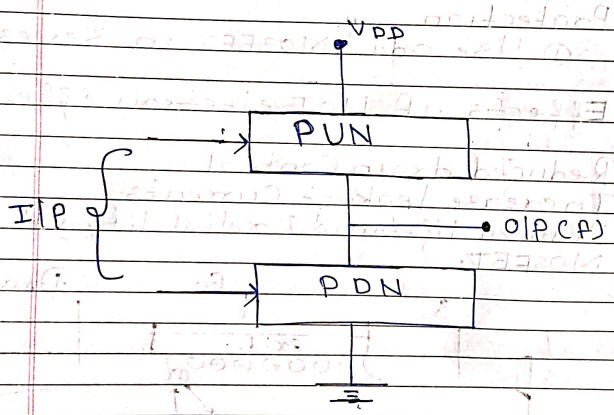
* Effects of HOT electron effects

① Reduced drain Conteal
② Increase leakage Currents.
③ Reduced lifetime & reliability of MOSFET.



* Designing Procedure of a Combin^{n} logic circuit using CMOS.

- CMOS is a comb^{n} of Two networks
- first is PUN, which is a pull-up Network.
- Second is PDN, which is Pull down network.
- Two Conn networks, PUN and PDN are Connected one after another



For: PDN ⟶ Nmosfet use
for: PUN ⟶ PMosfet use

* # Draw NAND, NOR, AND, OR gates using CMOS.

⟶

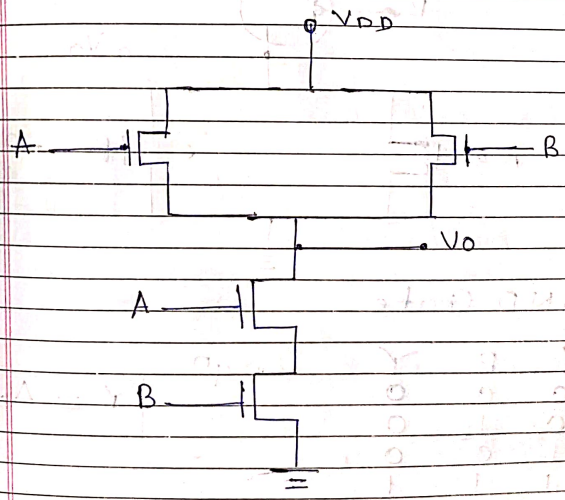
① NAND Gate

- Truth table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

eqn

$$Y = \overline{A.B}$$

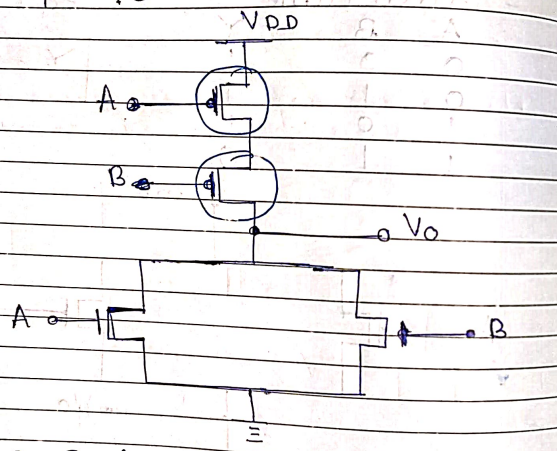


② NOR gate

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

eqn

$$Y = \overline{A+B}$$

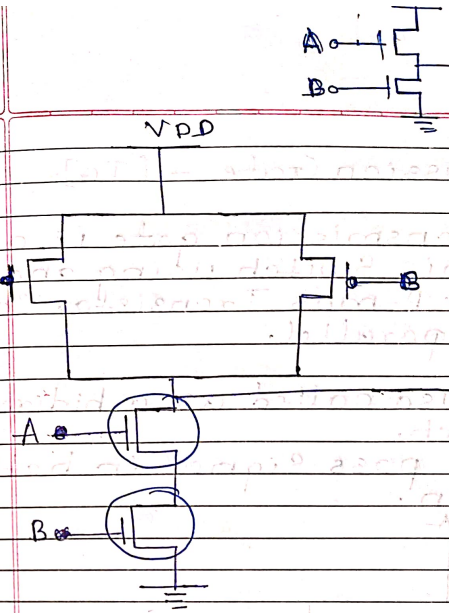


③ AND Gate

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

eqn

$$Y = A \cdot B$$

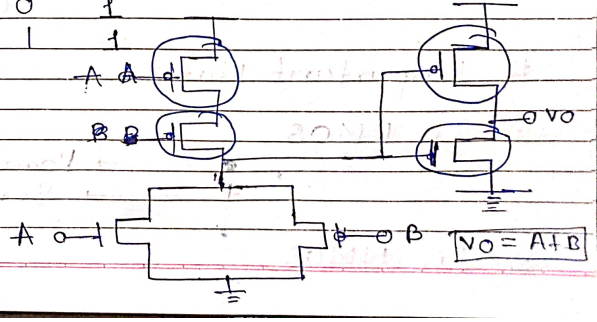


Inverter use only AND & OR

④ OR gate

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

$$Y = A+B$$

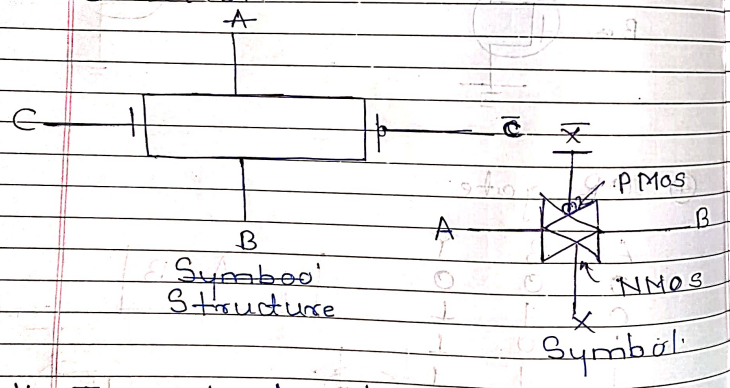


$V_O = A+B$

Transmission Gate - [TG].

① The Transmission gate is an electronic Switch using one NMos and one P-mos Transistor Connected in parallel.

- It is also called as a bidirectional Switch.
- it can pass Signal in both direction.



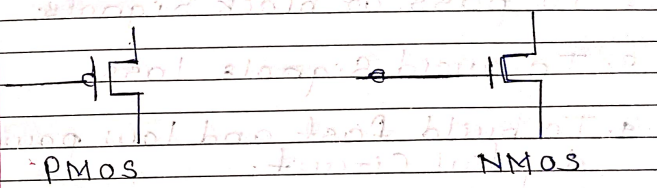
Important Note

① In PMOS

● ⟶ Parallel
+ ⟶ Series

② In NMOS

● ⟶ Series
+ ⟶ parallel



✱ Merits / Adv of Transmission Gate [TG].

- low Power Consumption
- passes both 0 and 1.
- High Switching Speed.
- Better Signal quality.
- High efficient.
- Bidirectional flow.

✱ Demerits / disadv of TG

① Need Complementary control Signal.
② more are req.
③ Design Complexity.
④ Not Suitable for high vtg application.

✱ Need

① A Transmission Gate is an elect

ronic Switch.

2. To pass or block Signals.

3. To avoid Signals loss.

4. To build fast and low power digitial Circuit.

5. To improve performance.